WORKED SOLUTION

# STEP II 2017, Question 6

Pure • Inequalities and Induction

**STEP II 2017, Question 6**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q6

## The key idea

Both parts compare the sum $S_n$ with $2\sqrt{n}$, which is roughly $\int_0^n x^{-1/2}\,\mathrm{d}x$. For the induction in (i), the step from $n$ to $n+1$ comes down to comparing $1/\sqrt{n+1}$ with $2(\sqrt{n+1}-\sqrt{n})$, and rationalising that difference settles it. In (ii), the smallest $C$ is the largest value of the gap between the right-hand side (without $C$) and $S_n$. So we study how that gap changes from $n$ to $n+1$, and the given inequality is exactly what tells us the gap always shrinks.

## Solution

### Part (i)

Let $P(n)$ be the statement $S_n \le 2\sqrt{n}-1$.

*Base case.* $S_1 = 1$ and $2\sqrt{1}-1 = 1$, so $P(1)$ holds (with equality).

*Inductive step.* Suppose $P(k)$ holds for some $k \ge 1$. Then

$$S_{k+1} = S_k + \frac{1}{\sqrt{k+1}} \le 2\sqrt{k} - 1 + \frac{1}{\sqrt{k+1}},$$

so it is enough to show that

$$2\sqrt{k} + \frac{1}{\sqrt{k+1}} \le 2\sqrt{k+1}, \qquad \text{that is,} \qquad \frac{1}{\sqrt{k+1}} \le 2(\sqrt{k+1}-\sqrt{k}).$$

Rationalising,

$$2(\sqrt{k+1}-\sqrt{k}) = \frac{2}{\sqrt{k+1}+\sqrt{k}} \ge \frac{2}{\sqrt{k+1}+\sqrt{k+1}} = \frac{1}{\sqrt{k+1}},$$

because $\sqrt{k} \le \sqrt{k+1}$ makes the denominator no bigger than $2\sqrt{k+1}$. Hence $S_{k+1} \le 2\sqrt{k+1}-1$, which is $P(k+1)$.

Since $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for every $k \ge 1$, $S_n \le 2\sqrt{n}-1$ for all positive integers $n$.

**Part (ii): the inequality**

For $k \geq 0$ both sides of $(4k+1)\sqrt{k+1} > (4k+3)\sqrt{k}$ are non-negative, and the left-hand side is positive, so the inequality is equivalent to the one between their squares. Now

$$\begin{aligned}(4k+1)^2(k+1) - (4k+3)^2 k &= (16k^2+8k+1)(k+1) - (16k^2+24k+9)k \\ &= (16k^3+24k^2+9k+1) - (16k^3+24k^2+9k) \\ &= 1 > 0.\end{aligned}$$

So $(4k+1)^2(k+1) > (4k+3)^2 k$, and taking positive square roots gives $(4k+1)\sqrt{k+1} > (4k+3)\sqrt{k}$ for all $k \geq 0$, as required.

**Part (ii): the smallest $C$**

Define

$$T_n = 2\sqrt{n} + \frac{1}{2\sqrt{n}} - S_n.$$

The inequality $S_n \geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - C$ is the same as $C \geq T_n$. So it holds for every $n$ exactly when $C$ is at least every $T_n$, and the smallest such $C$ is the largest value of $T_n$ (if it has one).

**Ask yourself** How do we find the largest $T_n$ without knowing a formula for $S_n$?

We look at the change $T_{n+1} - T_n$, in which $S_n$ disappears. Since $S_{n+1} - S_n = 1/\sqrt{n+1}$,

$$\begin{aligned}T_{n+1} - T_n &= 2\sqrt{n+1} - 2\sqrt{n} + \frac{1}{2\sqrt{n+1}} - \frac{1}{2\sqrt{n}} - \frac{1}{\sqrt{n+1}} \\ &= 2\sqrt{n+1} - 2\sqrt{n} - \frac{1}{2\sqrt{n+1}} - \frac{1}{2\sqrt{n}}.\end{aligned}$$

Multiply through by the positive quantity $2\sqrt{n}\sqrt{n+1}$:

$$\begin{aligned}2\sqrt{n}\sqrt{n+1}\,(T_{n+1} - T_n) &= 4(n+1)\sqrt{n} - 4n\sqrt{n+1} - \sqrt{n} - \sqrt{n+1} \\ &= (4n+3)\sqrt{n} - (4n+1)\sqrt{n+1}.\end{aligned}$$

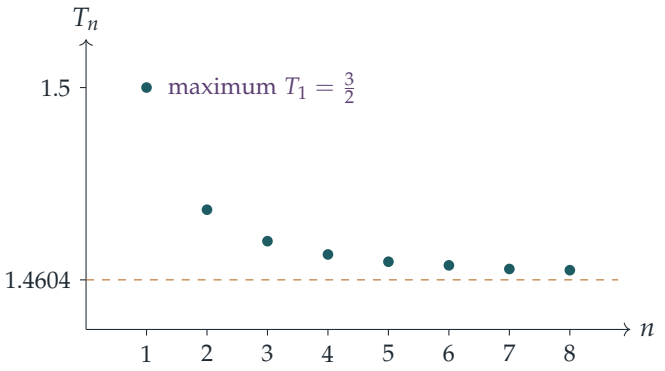
By the inequality we have just proved (with $k = n$), this is negative. Hence $T_{n+1} < T_n$ for every $n \geq 1$: the sequence $T_n$ is strictly decreasing, and its largest value is the first one,

$$T_1 = 2 + \tfrac{1}{2} - 1 = \tfrac{3}{2}.$$

So $S_n \geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - \frac{3}{2}$ for all $n \geq 1$. No smaller $C$ will do, because at $n = 1$ the inequality reads $1 \geq \frac{5}{2} - C$, which needs $C \geq \frac{3}{2}$. The smallest number is

$$C = \tfrac{3}{2}.$$

The first few values of $T_n$ show how fast it settles down:




(The vertical axis is stretched: it runs from about 1.45 to 1.51.)

## Checks

- For $n = 4$: $S_4 = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{2} \approx 2.7845$, which lies between $2\sqrt{4} + \frac{1}{4} - \frac{3}{2} = 2.75$ and $2\sqrt{4} - 1 = 3$, as both results require.
- A computer check of the first 200 000 values confirmed both bounds and that $T_n$ decreases; $T_n$ falls from 1.5 towards about 1.46035 and never returns to 1.5.
- The identity $(4k+1)^2(k+1) - (4k+3)^2 k = 1$ is easy to test at $k = 1$: $25 \cdot 2 - 49 = 1$.

## Common slips

- Answering with the limit of $T_n$ (about 1.46) instead of its maximum. A $C$ below $\frac{3}{2}$ fails at $n = 1$.
- Squaring an inequality without first noting that both sides are non-negative.
- Forgetting to show that $C = \frac{3}{2}$ is actually the smallest, by exhibiting the case of equality.

## Beyond STEP

The sequence $T_n$ decreases to a limit, and that limit is $-\zeta(\frac{1}{2}) \approx 1.46035$, where $\zeta$ is the Riemann zeta function, extended beyond the region where its defining series converges. Comparing a sum with an integral and then correcting by terms like $\frac{1}{2\sqrt{n}}$ is the start of the Euler–Maclaurin formula. The underlying fact that a bounded monotonic sequence converges is the monotone convergence theorem of first-year analysis; Abbott's *Understanding Analysis* treats it carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.